Worksheet

Stem-and-leaf plots 1

1 The ages of the first 30 people entering Milo National Park are listed below.
Use this information to complete parts **a** to **h**.

66	10	55	53	61	39	45	14	23	23
17	23	11	52	60	12	33	34	39	13
13	14	15	38	24	41	48	37	45	26

These are the steps required to organise this data into a stem-and-leaf plot:

- Give the plot a title
- Use the tens digit as the stem
- Use the ones digit as the leaf
- List the data in ascending order

Age of visitors to Milo National Park

Stem	Leaf
1	0 1 2 3 3 4 4 5 7
2	3 3 3 4 6
3	3 4 7 8 9 9
4	
5	
6	0 1 6

For example, the first entry 1 | 0 stands for '10' and the following entries stand for 11, 12, 13, 13, 14, etc.

a Complete the 4 and 5 stems on the above stem-and-leaf plot.

b What is the highest score?_____

c The **range** is the difference between the highest and lowest scores. What is the range?

_____

d How many people who entered the park were teenagers? What were the teenagers' ages?

_____

e How many people were over 45? _____

f Which age was most common? _____

g Which age group (stem) was most common? _____

h Which age groups were least common? _____

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2 The number of songs downloaded last weekend by a group of students is shown below.

No. of songs downloaded

Stem	Leaf
1	0 0 2 6
2	2 3 4 5
3	0 1 3 5 5 8
4	2 3 4 6 7
5	0 9
6	9

a How many students were surveyed? _____

b How many students downloaded fewer than 24 songs? _____

c What was the outlier score? _____

d What was the range? _____

e What percentage of people downloaded from 33 to 40 songs?

_____

f The **mode** is the most frequent score(s). What are the three modes for this data?

_____

3 The number of goals scored over a season by each player in a club is shown below.

Season goals

Stem	Leaf
0	7 8 9
1	1 2 4 5 5 5
2	2 3 3 6 7 8
3	0 0 9
4	4 4 5

a What was the mode? _____

b How many players in the club? _____

c How many scored more than 23 goals? _____

d What fraction of players scored between 10 and 22 goals? _____

e What was the range? _____

f What was the middle score? _____

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4 The numbers of people on board 20 buses over an afternoon were:

53	28	33	22	27	27	34	17	26	15
21	31	35	16	9	18	17	8	12	13

a Organise this data on a stem-and-leaf plot.

b What are the two modes? _____

c What is the outlier score? _____

d What is the smallest number of people on a bus? _____

e What percentage of buses had between 20 and 30 passengers?

_____

Answers

1 a

Stem	Leaf
1	0 1 2 3 3 4 4 5 7
2	3 3 3 4 6
3	3 4 7 8 9 9
4	1 5 5 8
5	2 3 5
6	0 1 6

b 66 **c** 56 **d** 6; 13, 14, 15, 17 **e** 7 **f** 23 **g** 10–19 **h** 50–59, 60–69

2 a 22 **b** 6 **c** 69 **d** 59 **e** 18% **f** 10, 35

3 a 15 **b** 21 **c** 9 **d** $\frac{2}{7}$ **e** 38 **f** 23

4 a

Stem	Leaf
0	8 9
1	2 3 5 6 7 7 8
2	1 2 6 7 7 8
3	1 3 4 5
4	
5	3

b 17 and 27 **c** 47 **d** 8 **e** 30%

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